Introduction to Computer Network

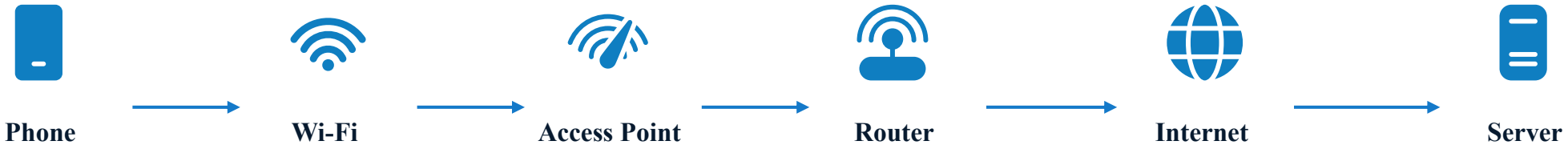
DSI102 – Introduction to Digital Technology
Week 5



Dr. Nantapong Keandoungchun

How did your phone reach this slide?

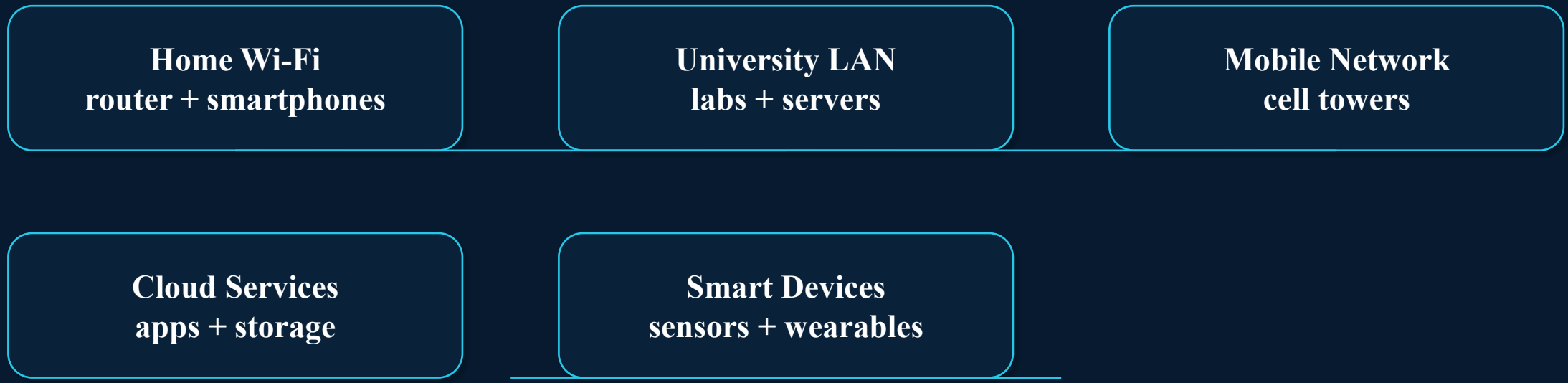
A quick network journey from student device to online content



Every request follows a chain of devices, links, and routing decisions.

Where networks are around us

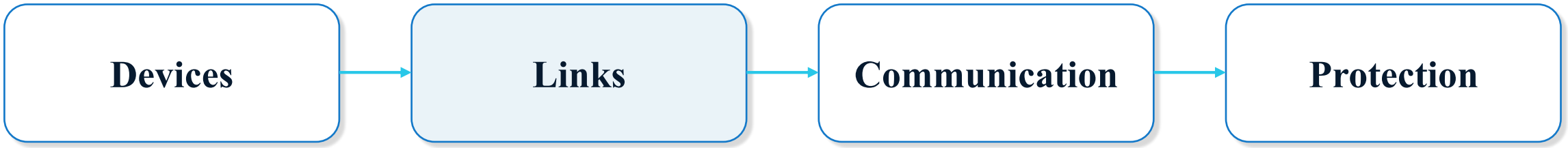
A network is not just “the Internet”; it is a connected system of devices.



What is a Computer Network?

Core definition from the Week 5 source deck

A link between two or more network devices.



Four questions of networking

Week 5 builds a mental model before protocol details.

What devices?
hub, switch, router, AP

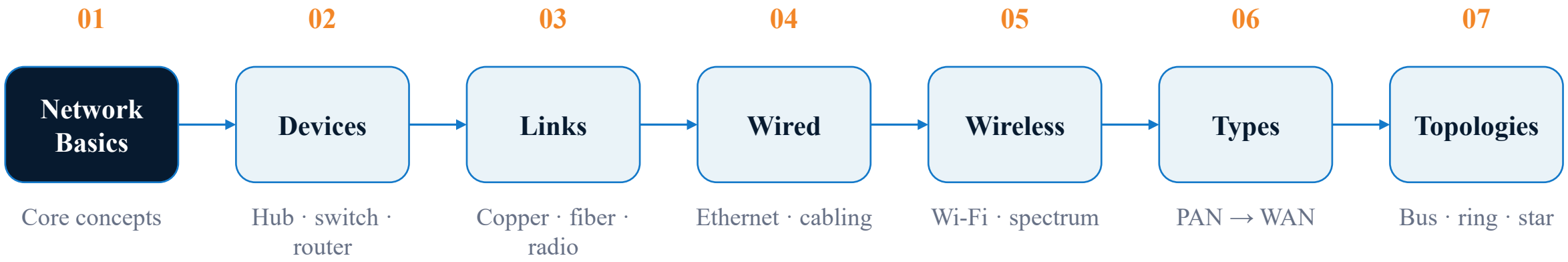
How connected?
wired or wireless links

How communicate?
frames, packets, addresses

How protected?
firewall and controls

Learning Journey

From devices to networks of networks



Week 5 moves from single devices to complete networks of networks.

01

Network Devices

The hardware roles that make connectivity possible



Network Device Ecosystem

Group the long device list into functional roles

Connect
Hub
Switch
Bridge

Route
Router
Gateway
Brouter

Access
Access Point
Modem

Extend
Repeater

Interface
NIC
Ethernet Card
Wi-Fi Card
RJ45

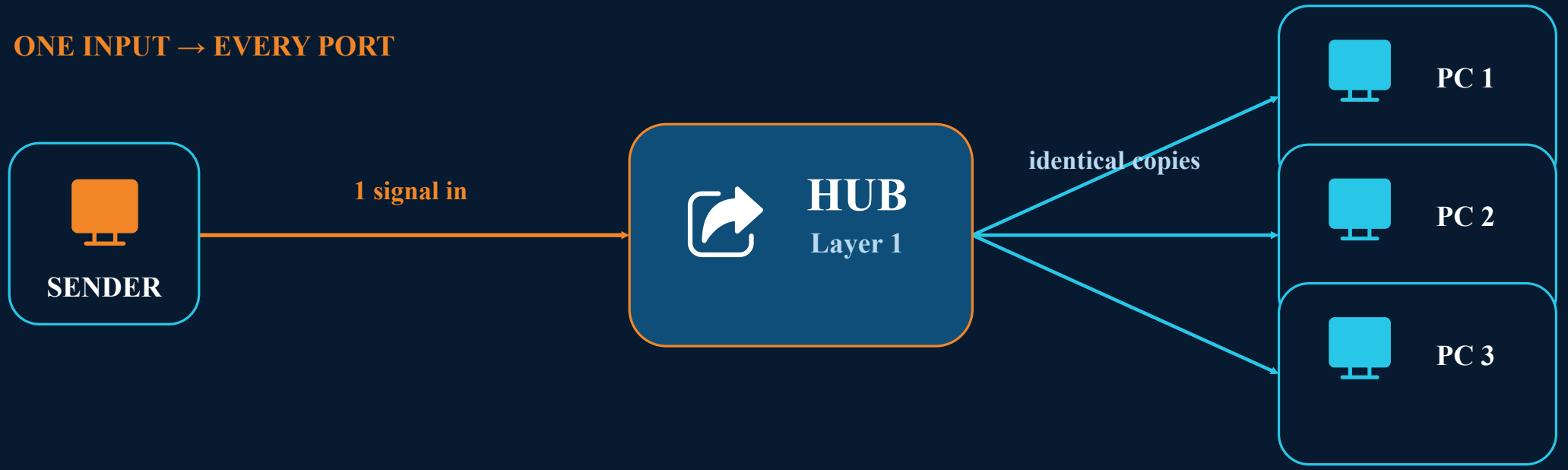
Protect / Control
Firewall
Load Balancer

Storage
NAS

Hub

A physical-layer device that broadcasts incoming signals

ONE INPUT → EVERY PORT



LAYER 1 DEVICE

Repeats electrical or optical signals.

NO ADDRESS INTELLIGENCE

Does not read MAC or IP addresses.

ONE COLLISION DOMAIN

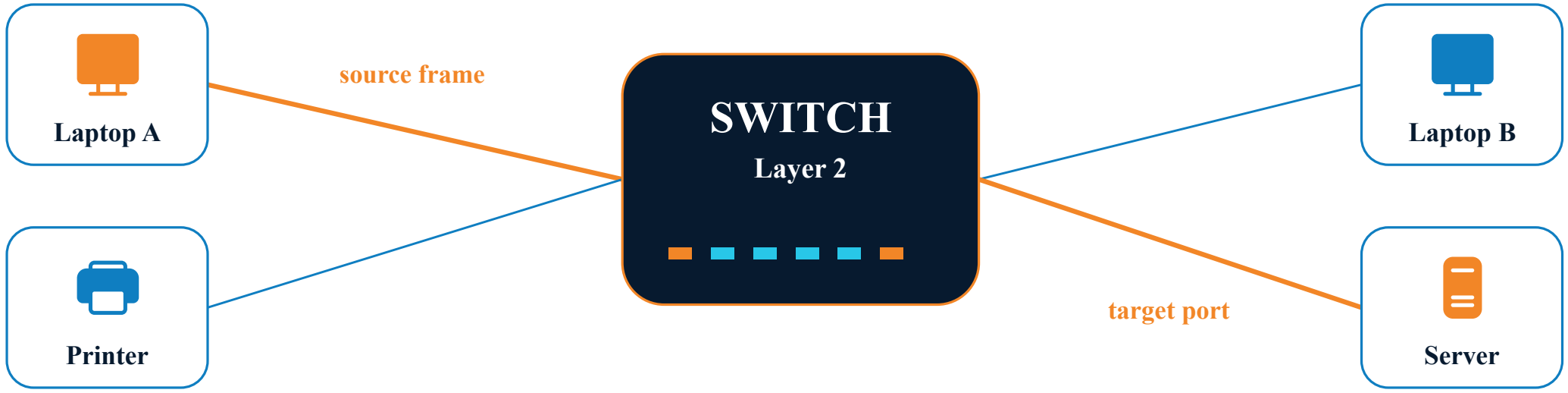
All connected devices share the medium.

Key idea: one input is repeated to every available port.

Switch

A data-link device that forwards frames using MAC addresses

LEARN → LOOK UP → FORWARD



1 LEARN SOURCE MAC

Record the port where the frame arrived.

2 LOOK UP DESTINATION

Check the MAC address table for the target.

3 FORWARD SELECTIVELY

Send the frame only through the target port.

The switch learns which MAC address is reachable through each port.

Router

A network-layer device that connects different networks



Read destination IP

Inspect the packet address before forwarding.

Choose the best route

Match the address against the routing table.

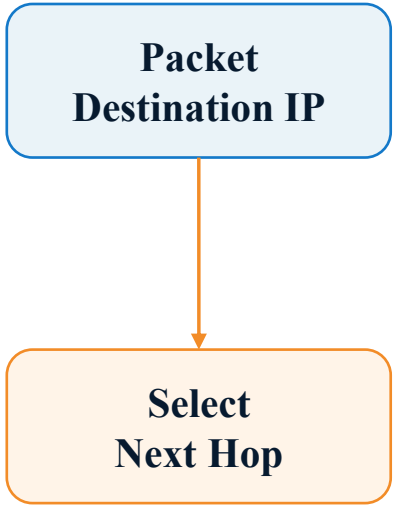
Connect networks

Forward traffic between different IP networks.

Routing Table

The router decides where to forward packets

Destination	Next Hop	Interface
192.168.1.0/24	Direct	LAN 1
10.0.0.0/24	Direct	LAN 2
0.0.0.0/0	ISP Router	WAN



Longest prefix wins

The most specific matching route is selected.

Default route

0.0.0.0/0 handles destinations with no closer match.

Direct connection

“Direct” means the destination network is attached.

Modem

Modulator / Demodulator



Converts signals between service-provider media and digital devices.

Types of Modem

Examples from the source deck



Optical Modem



Digital Modem



Cable Modem



Satellite Modem

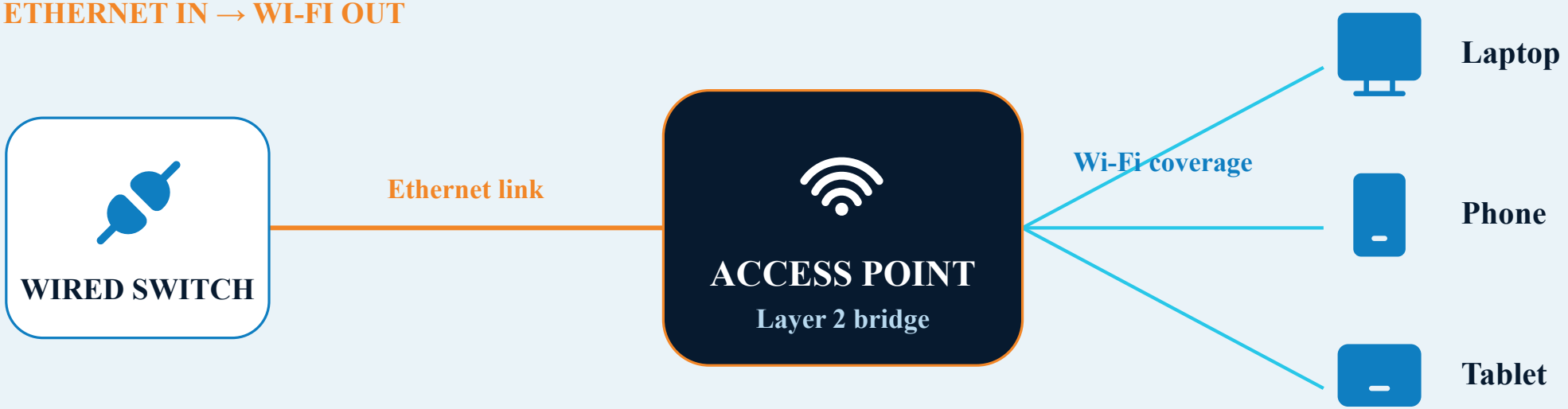


Dial-up Modem

Access Point

Bridges wireless clients into a wired network

ETHERNET IN → WI-FI OUT



EXTENDS COVERAGE

Connects wireless clients across the local area.

BRIDGES AT LAYER 2

Moves frames between wired and wireless media.

KEEPS ONE SUBNET

Clients remain in the same local IP network.

One access point connects many wireless devices to the wired LAN.

Repeater

Extends a network by regenerating a weak signal



- Useful when distance weakens signal quality
- Repeats or regenerates traffic for the next segment

Hub vs Switch vs Router

Three roles, three layers, three forwarding ideas

HUB
Layer 1
Broadcast
No address intelligence

SWITCH
Layer 2
MAC address
Inside a LAN

ROUTER
Layer 3
IP address
Between networks

Quick Check: Which device?

Choose hub, switch, router, AP, modem, or repeater

- Connects two different networks using IP addresses
- Repeats an incoming signal to all ports
- Lets Wi-Fi devices join the wired LAN
- Converts provider signal to digital network use

Discuss in pairs for 90 seconds.

02

Network Links

How devices connect and exchange signals

What is a Link?

A communication path that connects network devices



Two Families of Links

Wired and wireless media carry signals differently

WIRED
Coaxial
Twisted Pair
Fiber Optic

WIRELESS
Radio
Microwave
Satellite
Cellular
Infrared

03

Wired Networks

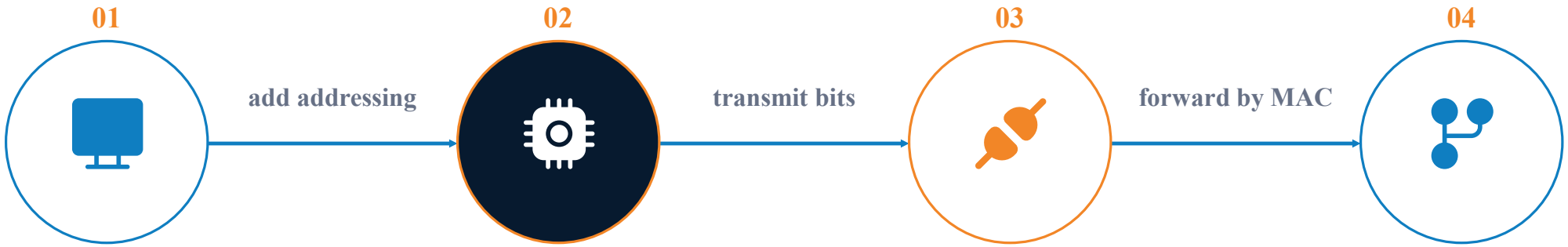
Copper and fiber links in physical network infrastructure



Ethernet at a Glance

A common LAN path from device to switch

DATA BECOMES AN ETHERNET FRAME



DEVICE

Creates the data

NIC / CARD

Adds source + destination MAC

RJ45 LINK

Carries the Ethernet frame

SWITCH

Selects the destination port

ONE CABLE PER PORT

Each switch port forms a dedicated link.

FRAME IN TRANSIT



SPEED LADDER

Fast Ethernet → Gigabit Ethernet and beyond.

Ethernet carries addressed frames over a dedicated link to a switch.

RJ45 Connector

Common connector form for Ethernet twisted-pair cabling

Key points
- 8-position connector
- used with twisted-pair Ethernet
- wiring follows T568A or T568B



T568A Wiring Standard

8-pin wiring order shown with editable labels



T568B Wiring Standard

8-pin wiring order shown with editable labels



T568A vs T568B

The orange and green pairs swap positions

T568A
Pins 1–2: Green pair
Pins 3 & 6: Orange pair

T568B
Pins 1–2: Orange pair
Pins 3 & 6: Green pair

Straight-through Cable

Same wiring standard at both ends



Also valid: T568A to T568A

Crossover Cable

Different wiring standards at the two ends



Source examples: computer–computer, router–router, switch–switch.
Modern context: many devices can auto-detect cable type.

RJ11 vs RJ45

Telephone interface vs Ethernet network connector

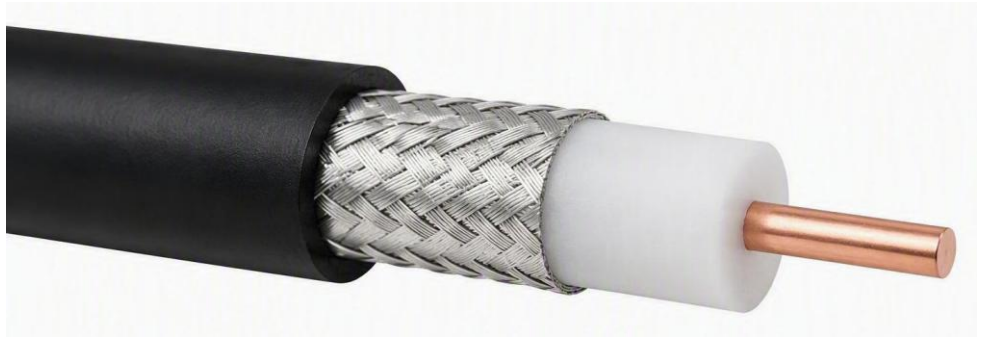
RJ11
Telephone connections
Fewer contacts / wires
Smaller connector

RJ45
Ethernet cabling
8P8C form
Larger connector

Coaxial Cable

A layered cable with center conductor and shielding

OUTSIDE → CENTER



FOUR CONCENTRIC LAYERS

01 **OUTER JACKET**
outermost layer

02 **SHIELD**
surrounds the dielectric

03 **DIELECTRIC**
surrounds the conductor

04 **INNER CONDUCTOR**
center core

WHERE IT APPEARS

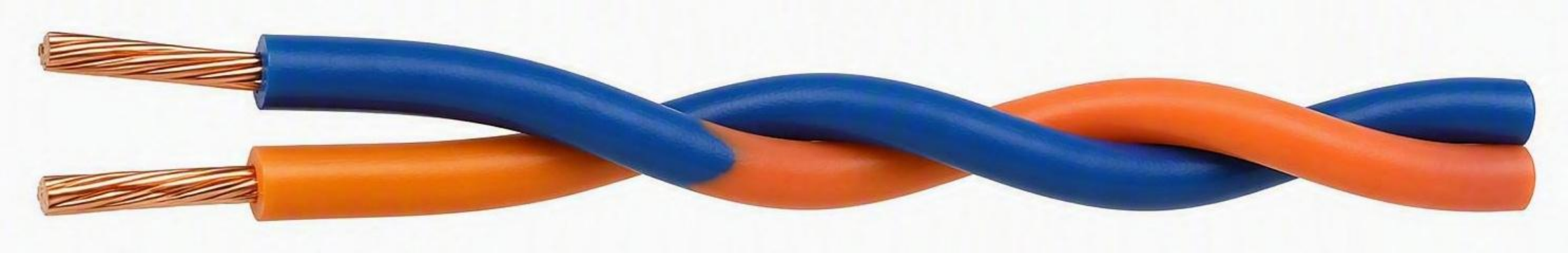
Cable television and legacy communication contexts

WHY THE SHIELD MATTERS

Designed to reduce surrounding interference

Twisted-pair Cable

Pairs of conductors are twisted to improve electromagnetic compatibility



Twisting keeps both wires equally exposed to interference.

WHY IT WORKS

1 Positions alternate

Each conductor repeatedly moves closer to—and farther from—noise sources.

2 Noise becomes common

Interference is induced similarly in both wires instead of mainly in one.

3 The receiver subtracts

Common noise cancels, leaving the intended difference signal clearer.

Shielding Terminology

Source abbreviations used in twisted-pair cable tables



U

Unshielded

F

Foil shielding

S

Screen / braid

TP

Twisted pair

Ethernet Cable Categories I

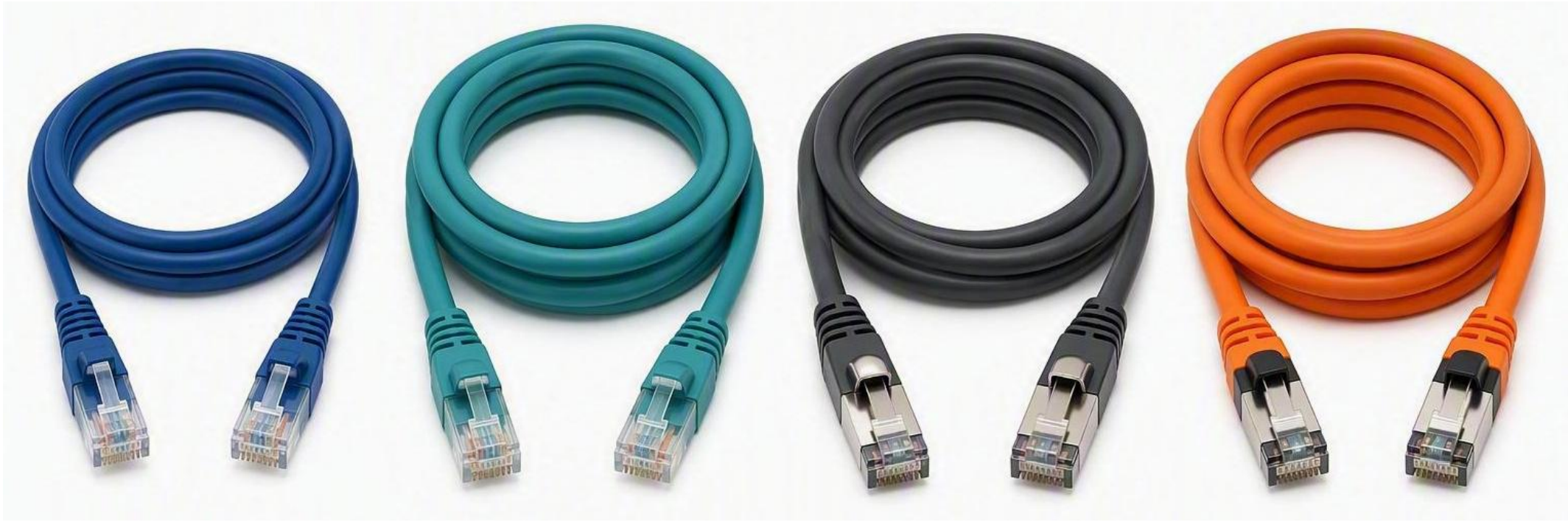
Earlier to common LAN categories



CAT 3	CAT 5	CAT 5e
16 MHz	**100 MHz**	**100 MHz**
10BASE-T · 100BASE-T4	100BASE-TX · 1000BASE-T	1000BASE-T · 2.5GBASE-T

Ethernet Cable Categories II

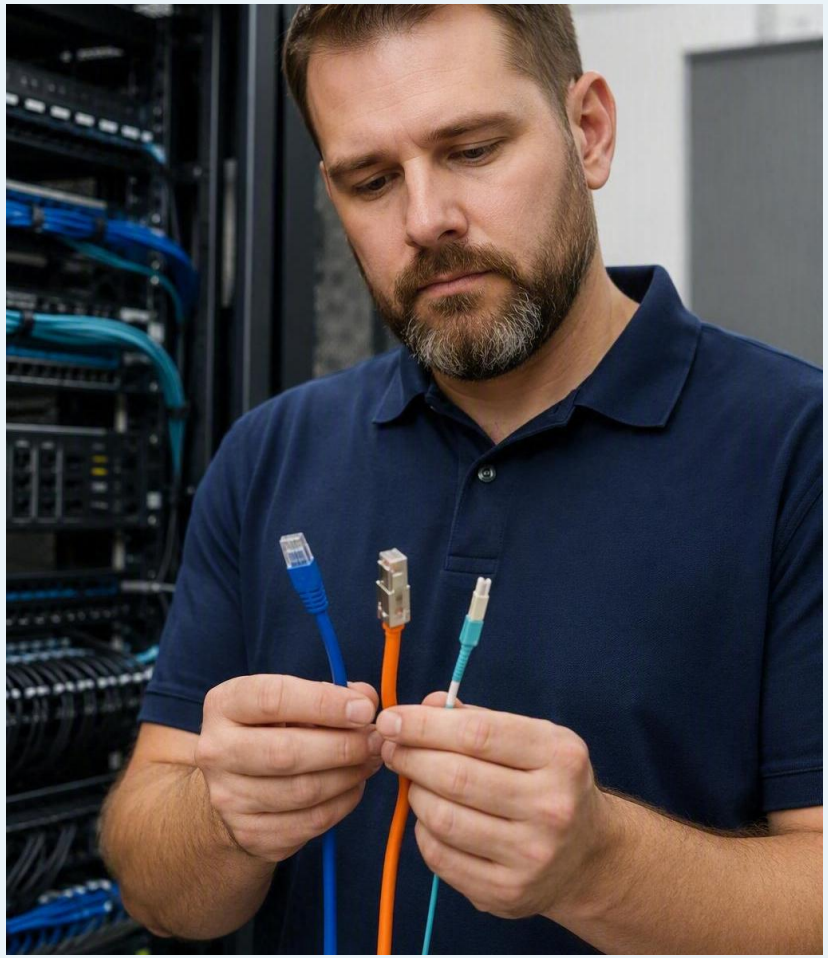
Higher-performance copper cabling



CAT 6	CAT 6A	CAT 7 / 7A	CAT 8
250 MHz	**500 MHz**	**600 MHz / 1 GHz**	**2 GHz**
5GBASE-T · 10GBASE-T	10GBASE-T	special connectors	25GBASE-T · 40GBASE-T

Choosing a Copper Cable

Match medium to environment and performance need



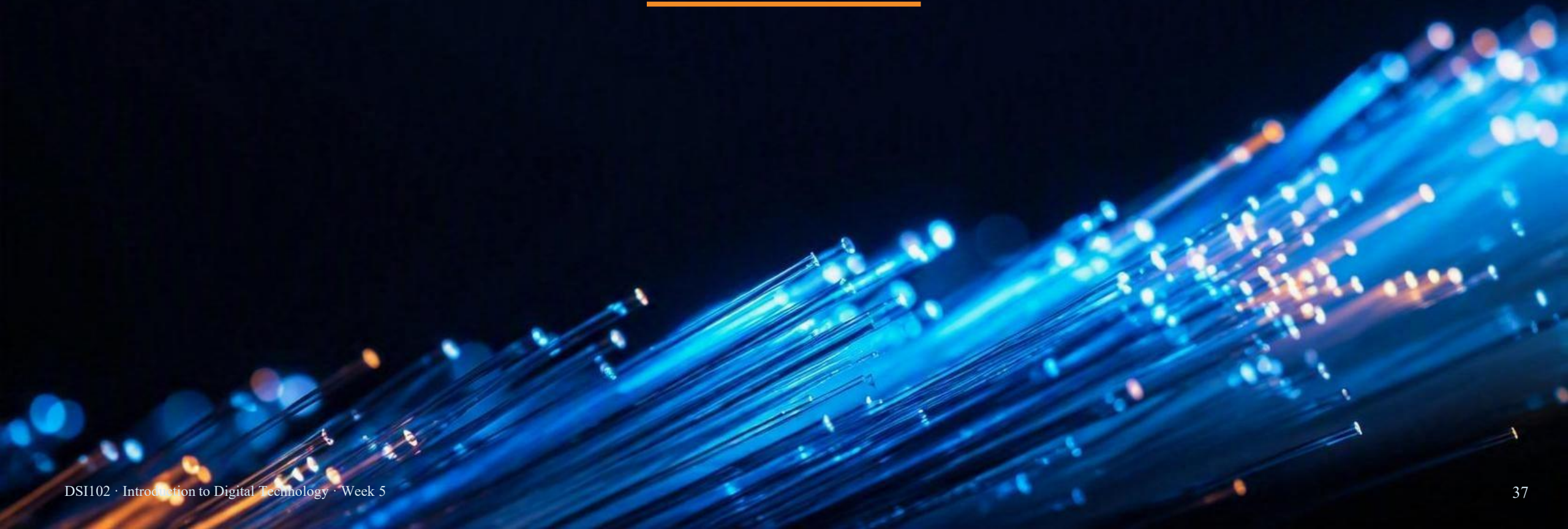
START WITH THE USE CASE		RECOMMENDATION
01	**Home / basic LAN**	common twisted pair
02	**Office LAN**	Cat 5e / Cat 6 context
03	**10G short run**	higher category copper
04	**High-performance install**	consider shielding / fiber

Choose for the environment first—then the speed target.

04

Fiber Optic Networking

Using light to carry data across distance



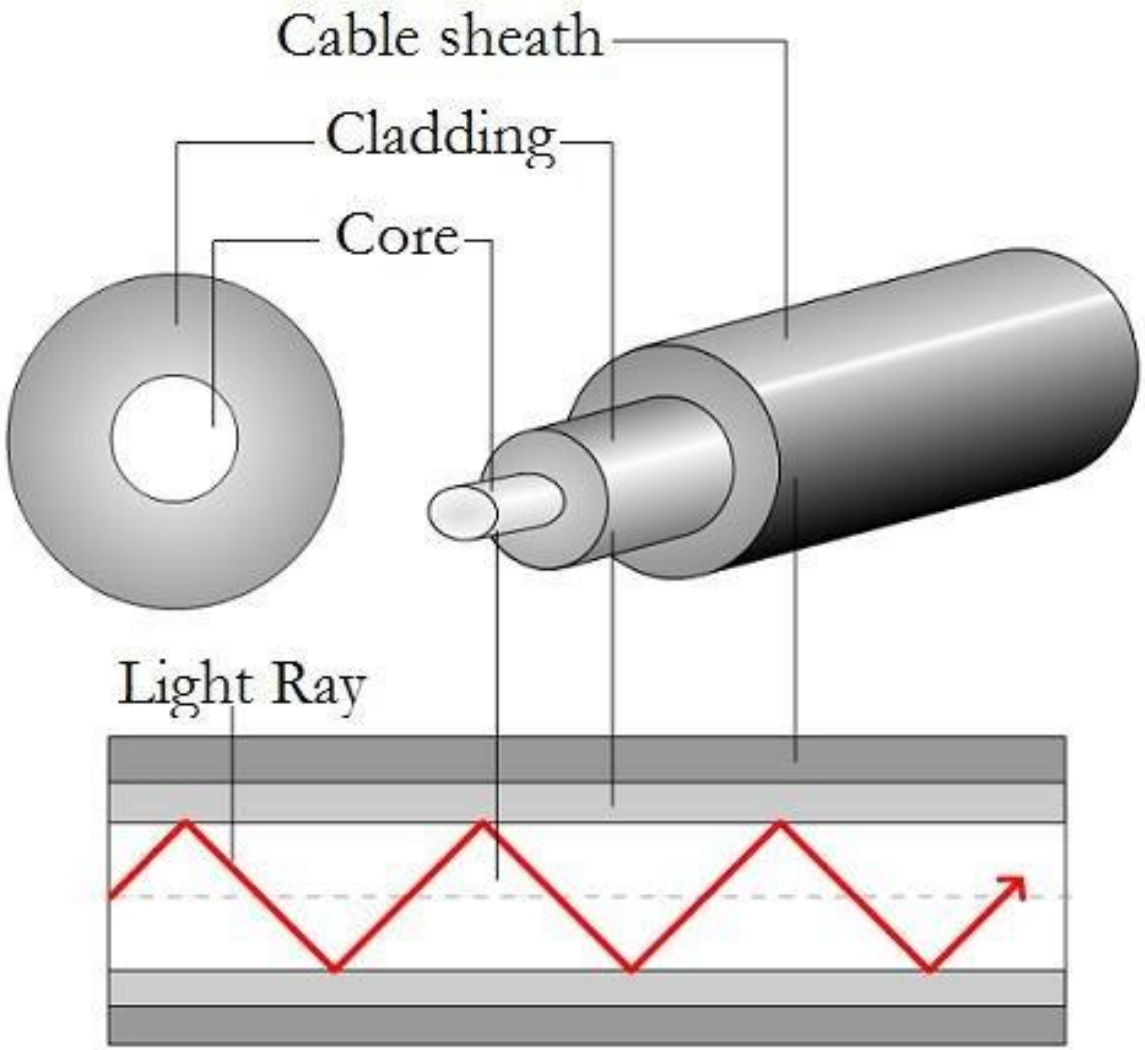
What is Fiber Optic Cable?

Fiber carries information using light.

Electrical data
→ light signal
→ fiber path
→ data receiver

Anatomy of Fiber

Core, cladding, buffer, and outer jacket

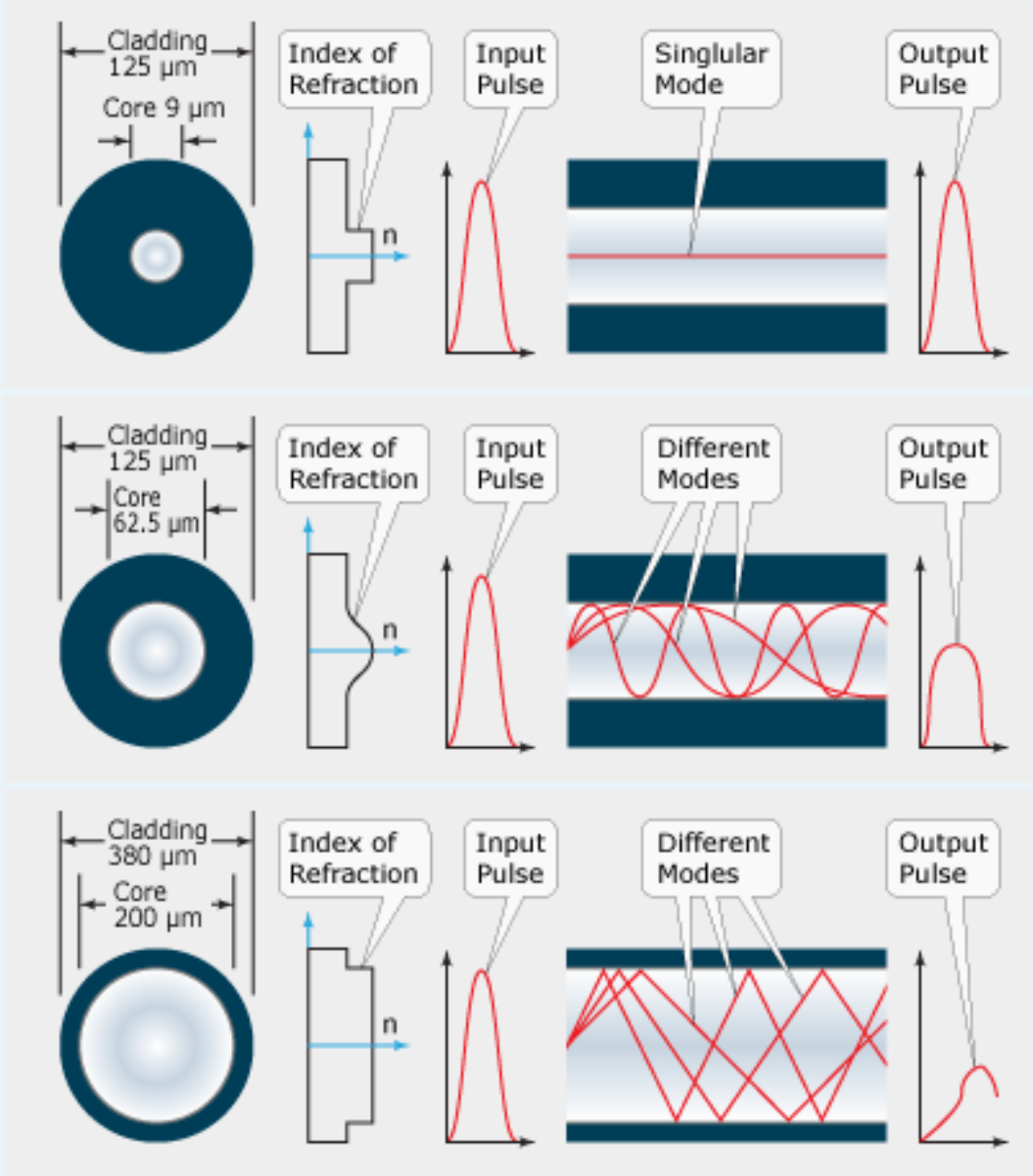


Fiber Types

Source classifications and high-level distinction

Single-mode
Step-index single-mode
Long-distance backbone

Multi-mode
Graded-index multi-mode
Step-index multi-mode
Shorter-distance environments



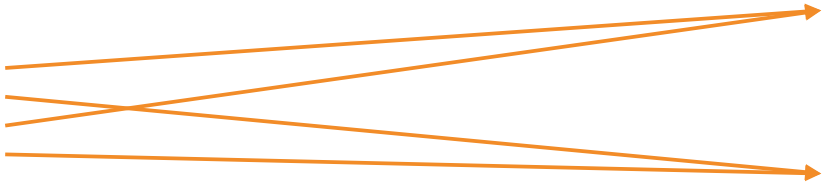
Single-mode vs Multi-mode

One path vs multiple paths of light

Single-mode
One narrow light path
Long-distance telecom/backbone
Requires laser-based transmission

Multi-mode
Multiple light paths
Building or campus distances
Can use less powerful light sources





Fiber Optic Connectors

Examples from the source deck

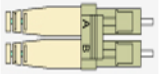
FC
Screw-on

LC
Snap-on

MTRJ
Snap-on

SC
Snap-on

ST
Twist-on

Connector Type	Picture	Coupling Type	# of Fibers	Application
FC		Screw On	1	Telecommunications
LC		RJ45 Style Snap On	1 or 2	Gigabit Ethernet
MTRJ		RJ45 Style Snap On	2	Gigabit Ethernet
SC		Snap On	1	CATV
ST		Twist On	1	LANS

05

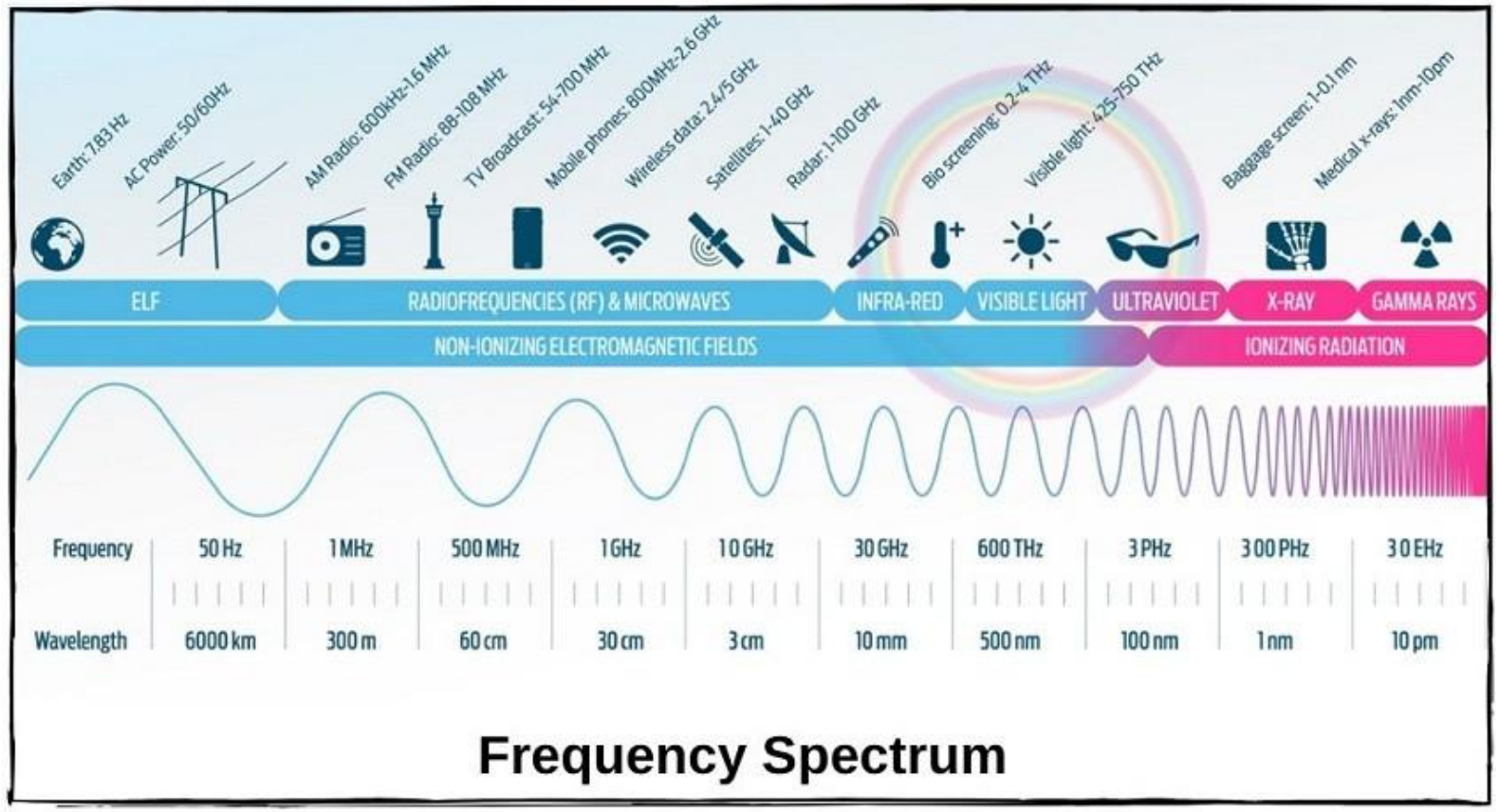
Wireless Networks

Connecting devices without a physical cable



Wireless Communication

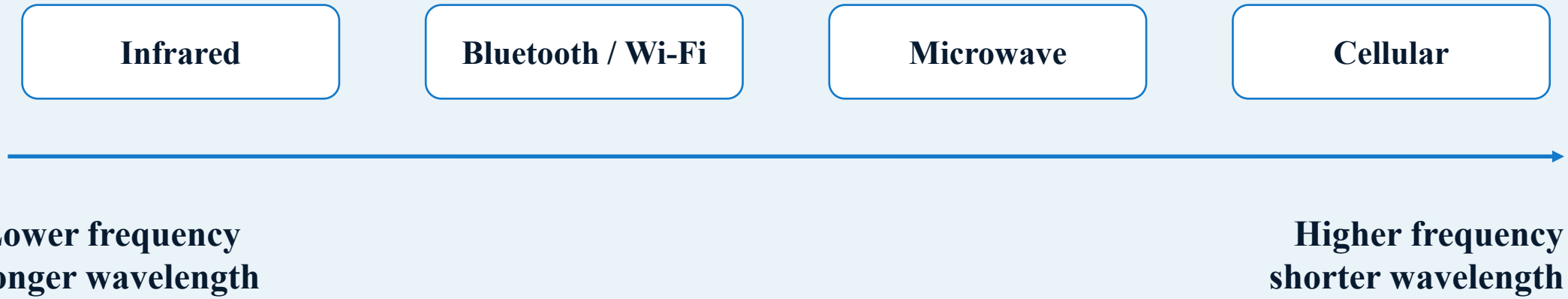
Signals travel through space instead of cable

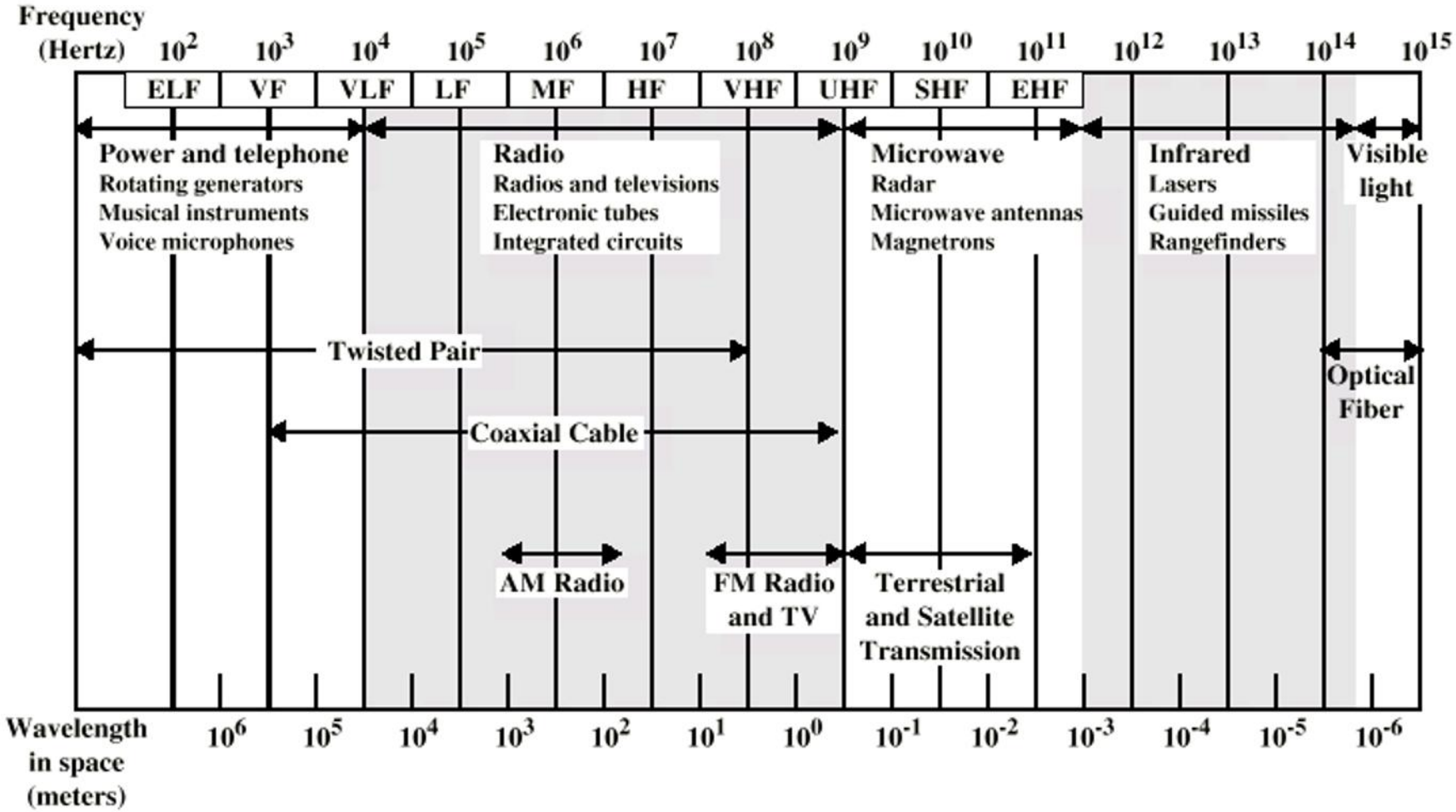


Wireless links include Wi-Fi, Bluetooth, microwave, satellite, cellular, and infrared.

Frequency

Higher frequency means shorter wavelength in a conceptual spectrum

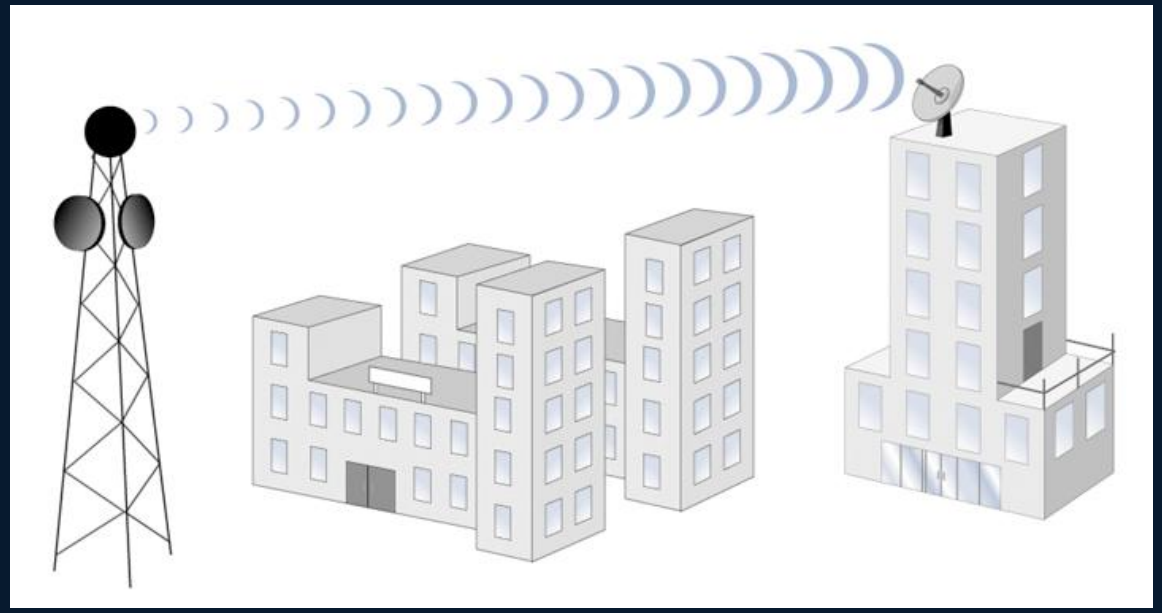




ELF = Extremely low frequency
VF = Voice frequency
VLF = Very low frequency
LF = Low frequency
MF = Medium frequency
HF = High frequency
VHF = Very high frequency
UHF = Ultrahigh frequency
SHF = Superhigh frequency
EHF = Extremely high frequency

Terrestrial Microwave

Tower-to-tower line-of-sight transmission



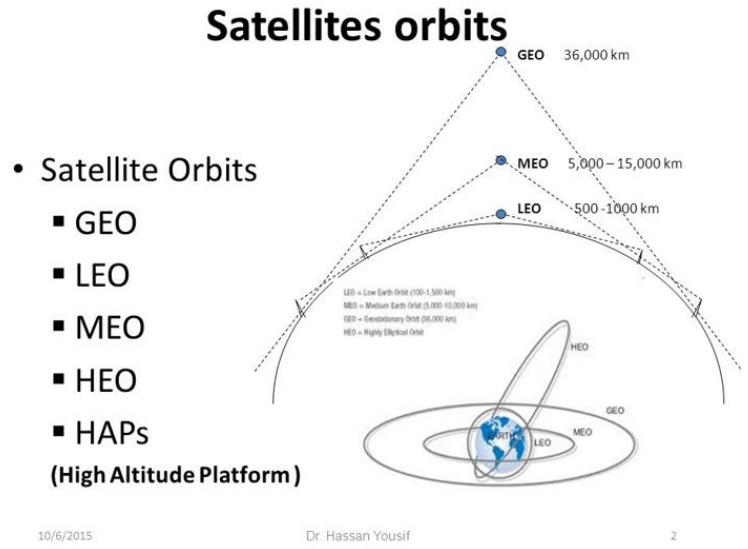
- Approximately 20–30 miles between towers
- Line-of-sight: signals cannot pass through solid objects
- Source example: 100 Mb per second

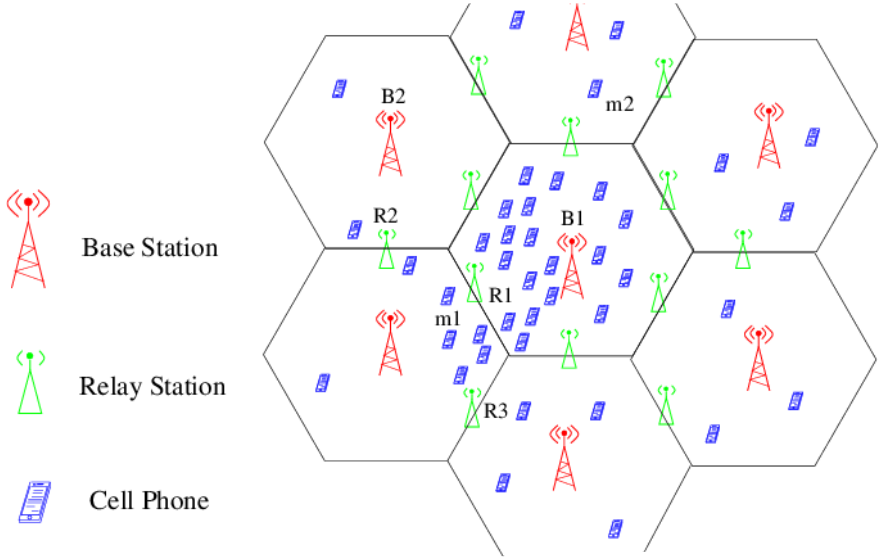
Satellite and Cellular Networks

Wide-area coverage uses different wireless infrastructure

Satellite Microwave
Ground station → satellite → ground station
Long-distance coverage

Cellular Networks
City divided into cells
Each cell has a transmission tower





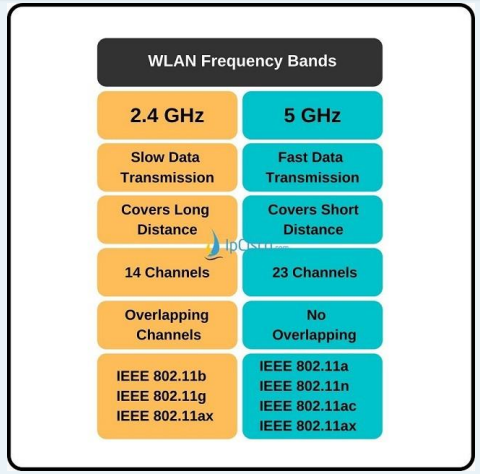
Short-range Wireless Examples

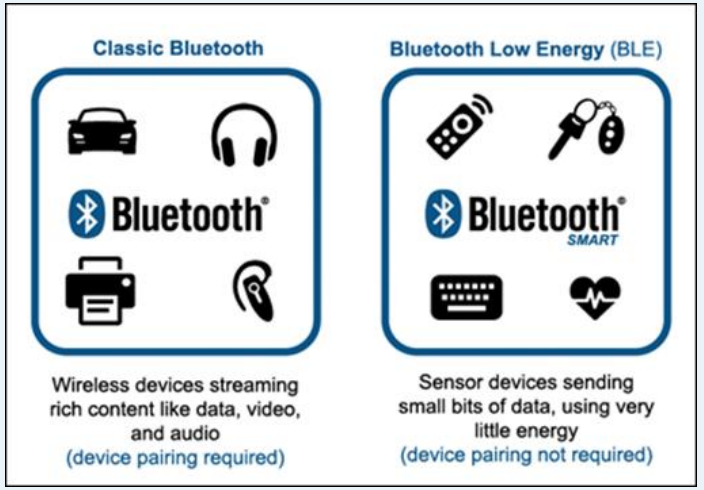
Wireless LAN, Bluetooth, and infrared

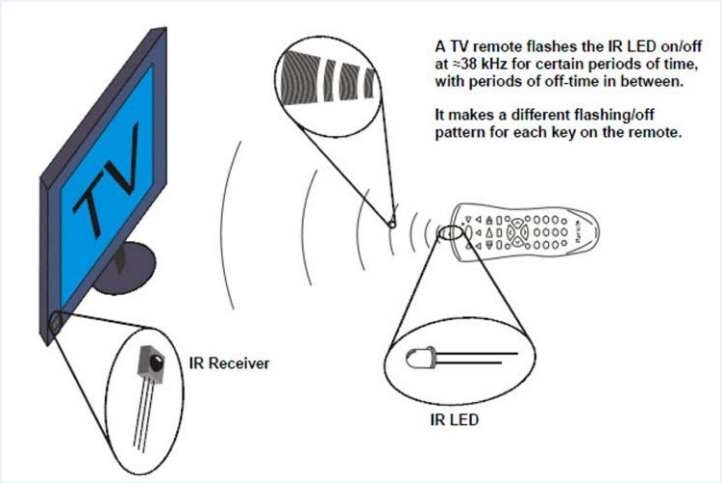
Wireless LAN
Wi-Fi bands and channels connect local devices.

Bluetooth
Short-range personal device connectivity.

Infrared
Line-of-sight communication, sensors, and remotes.

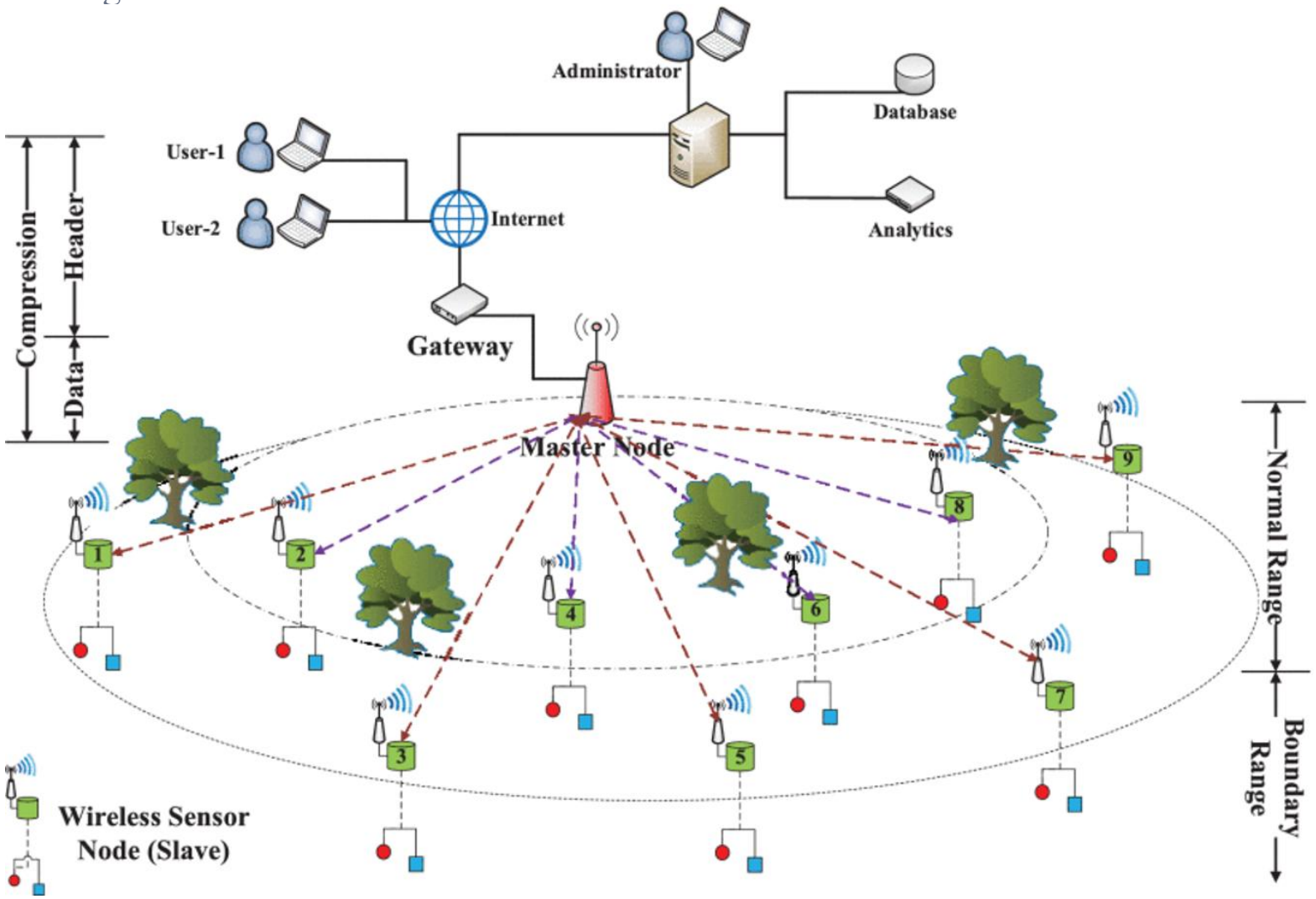






Wireless Sensor Network

Multiple sensors send readings to a central location



06

Network Types

Classifying networks by scale and purpose



Networks by Geographic Scope

From personal devices to global connectivity



Other Network Types

Purpose-based and overlay network classifications

SAN
Dedicated storage network

EPN
Private enterprise network

VPN
Private overlay on public network

Cloud Network
WAN-like infrastructure delivered by cloud services

Intranet, Internet, Extranet, Darknet

Different access models and boundaries

Intranet
controlled by one organization
authorized users

Extranet
connects to external partners
shared resources

Internet
global system of connected networks

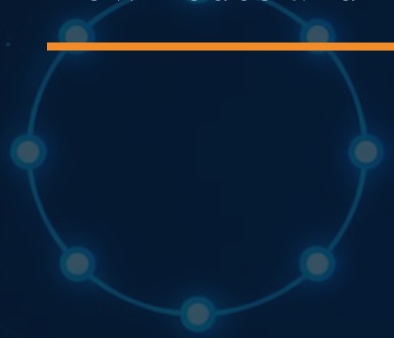
Darknet
accessed through specialized software and protocols

07

Network Topology

How nodes and links are arranged







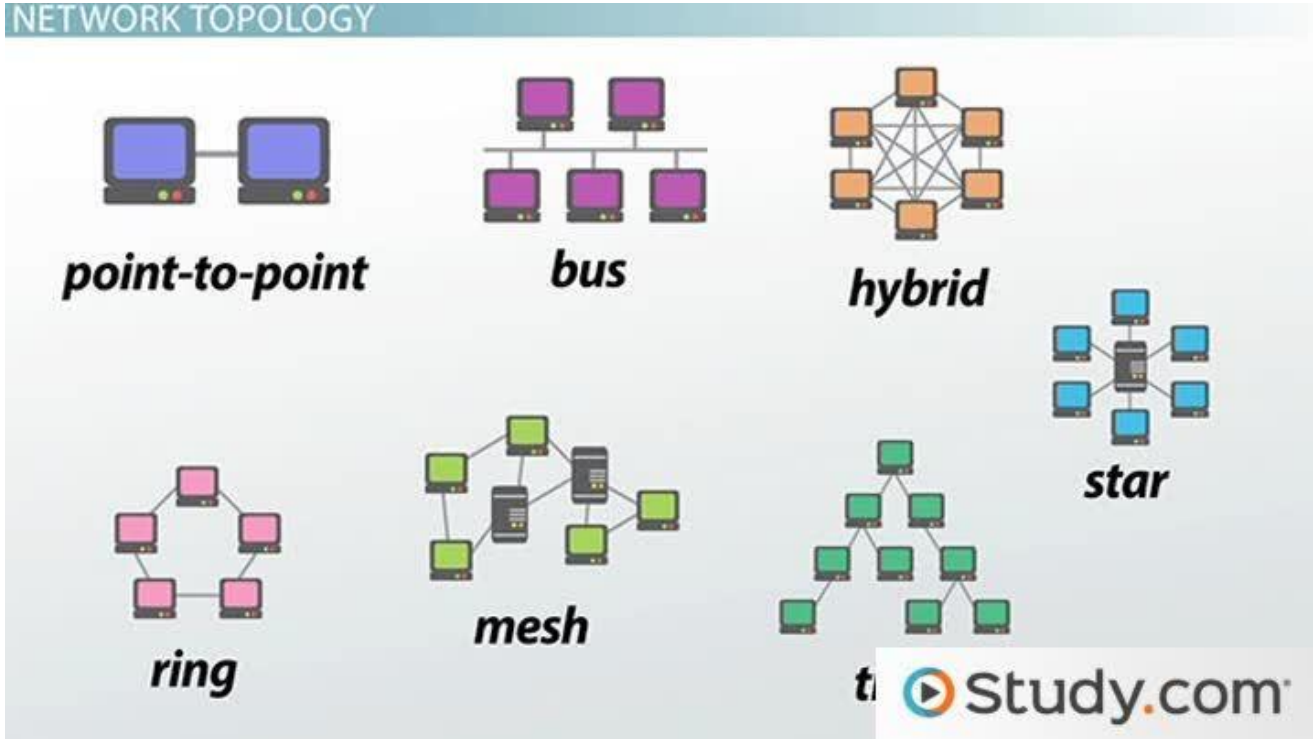




What is Network Topology?

The physical or logical arrangement of network nodes and links

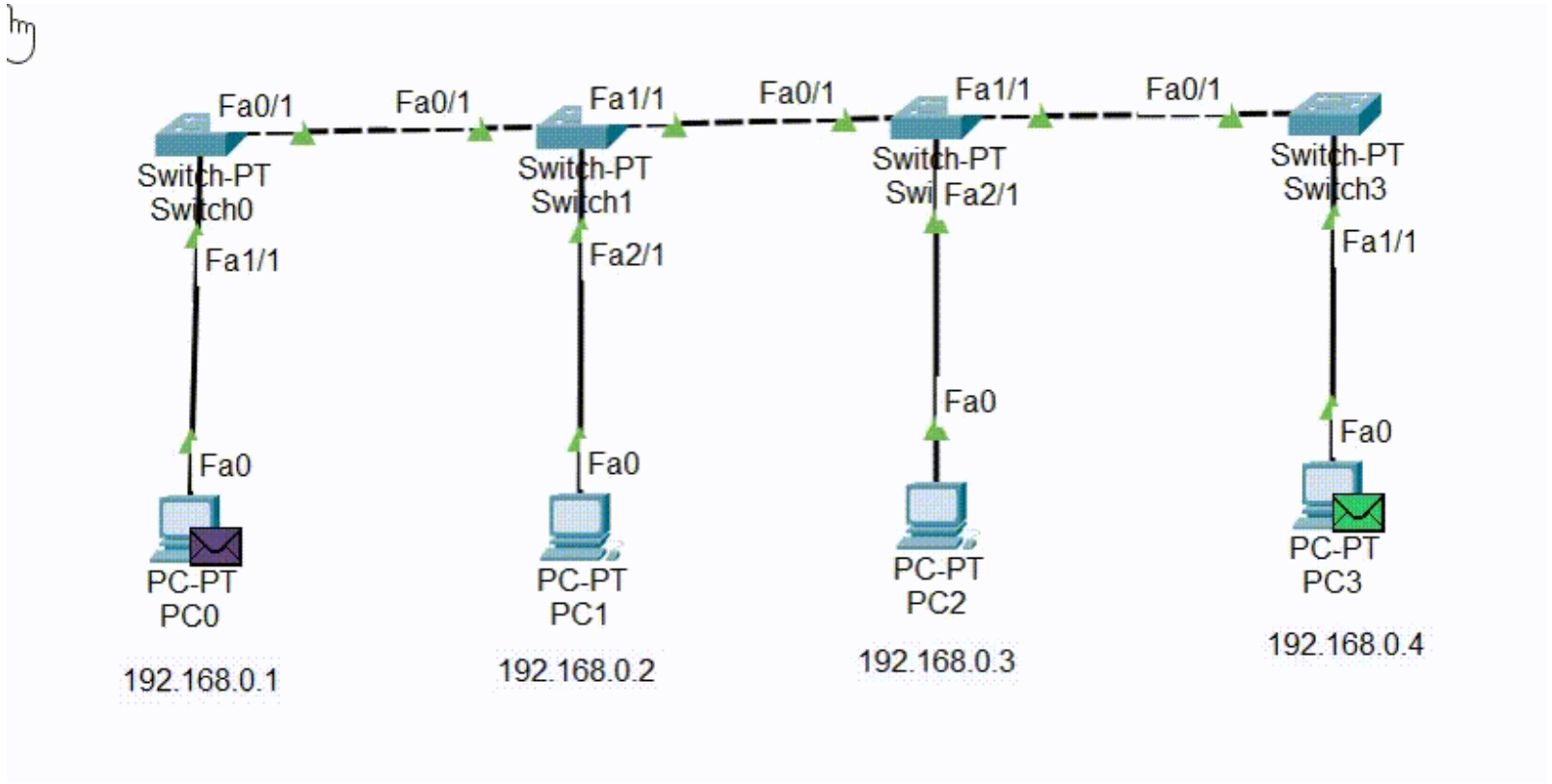
Topology describes how devices are connected and how traffic may flow.



Topology Comparison

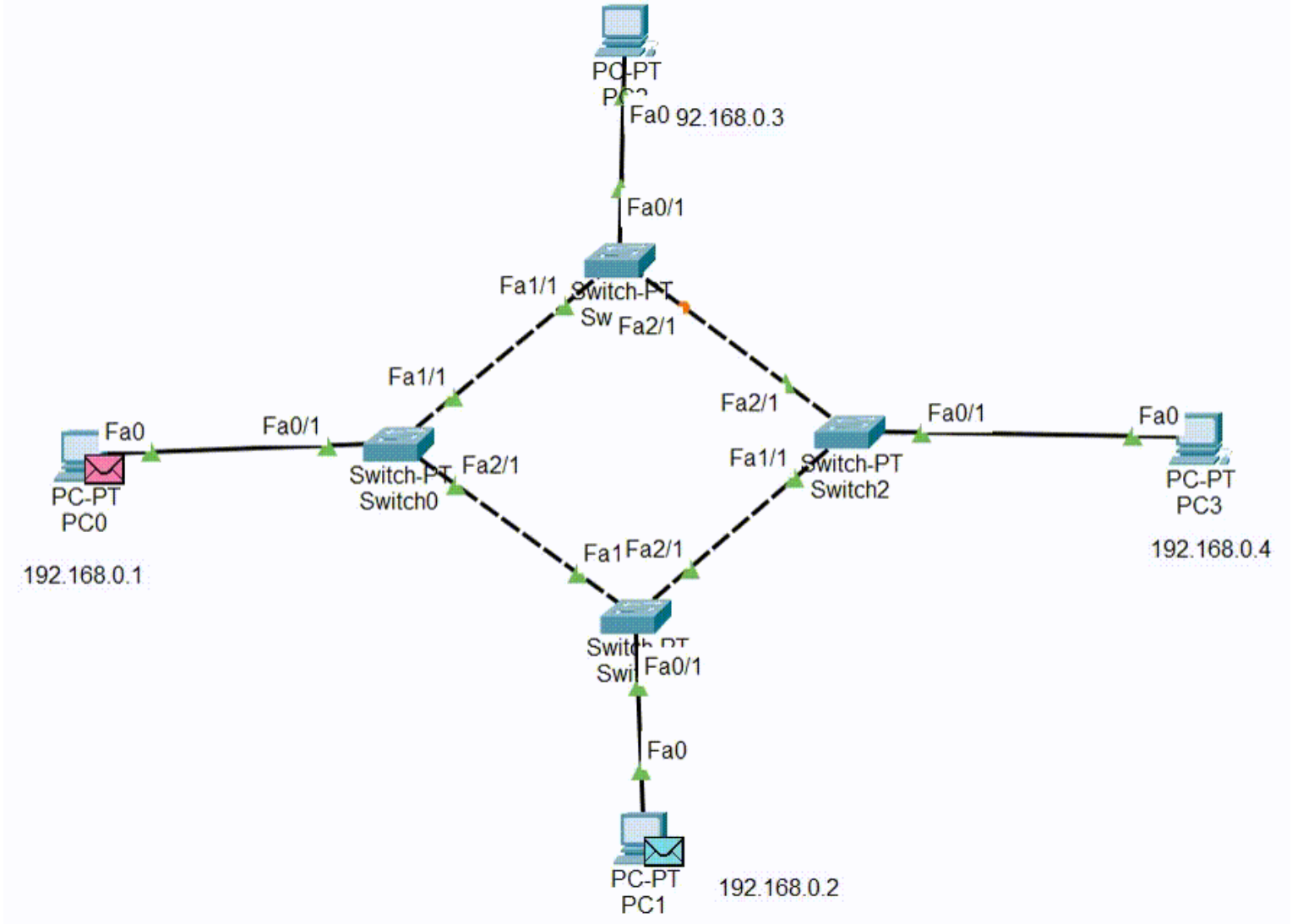
Structure, fault tolerance, and complexity

Topology	Structure	Fault Tolerance	Complexity
Bus	shared backbone	low	low
Ring	closed loop	moderate	moderate
Star	central node	central point risk	low–moderate
Mesh	many paths	high	high
Tree	hierarchy	depends on root/core	moderate



Bus Topology

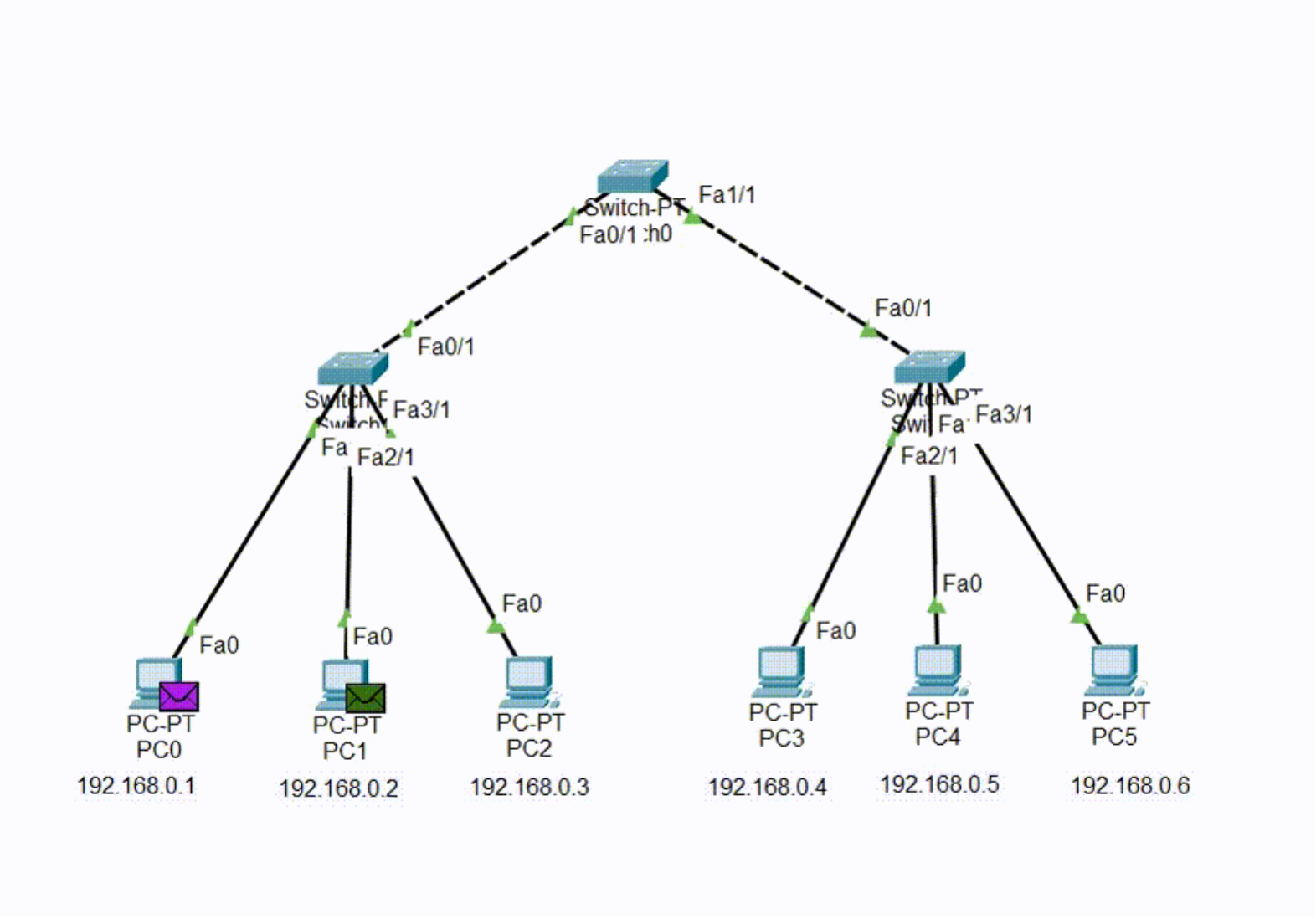
https://www.geeksforgeeks.org/implementation-of-bus-topology-in-cisco/



Ring Topology

PC0 to PC2 and PC1 to PC3

https://www.geeksforgeeks.org/implementation-of-ring-topology-in-cisco/?ref=ml_lbp

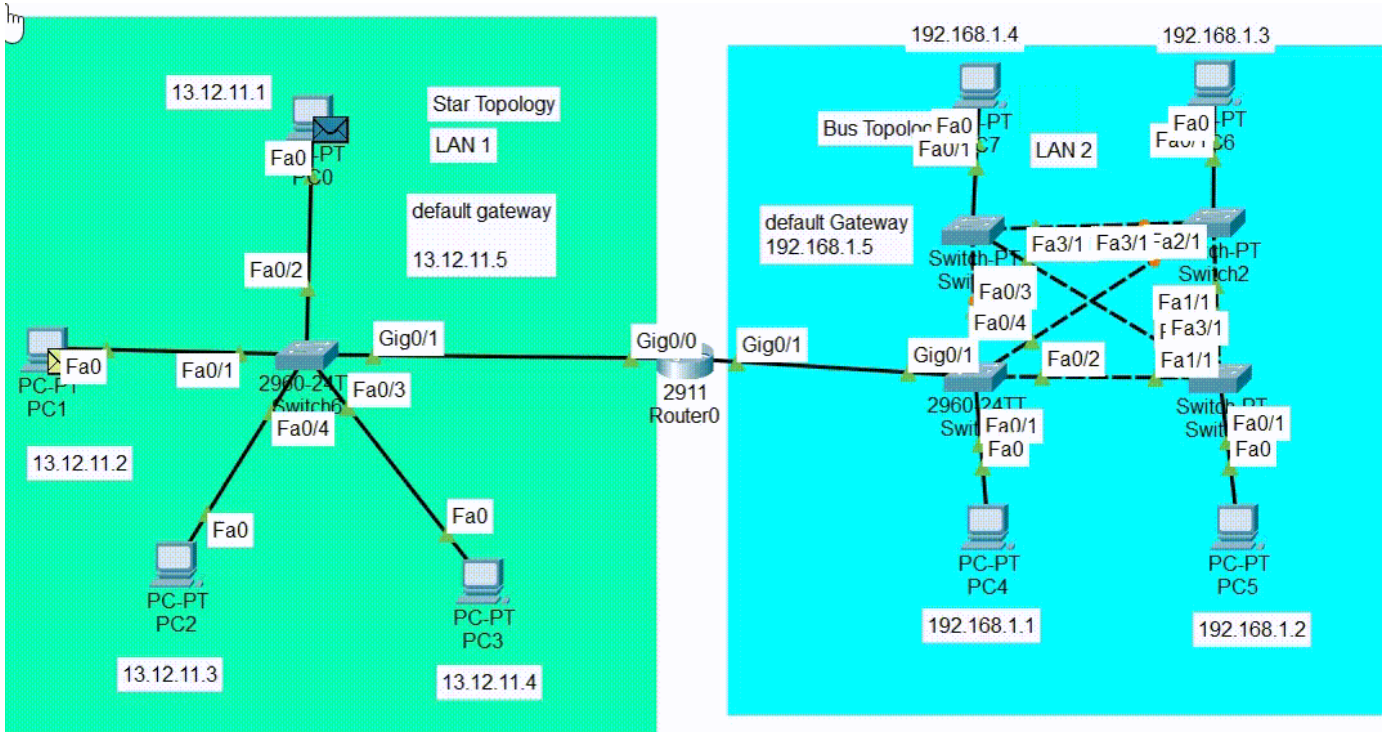


Tree Topology

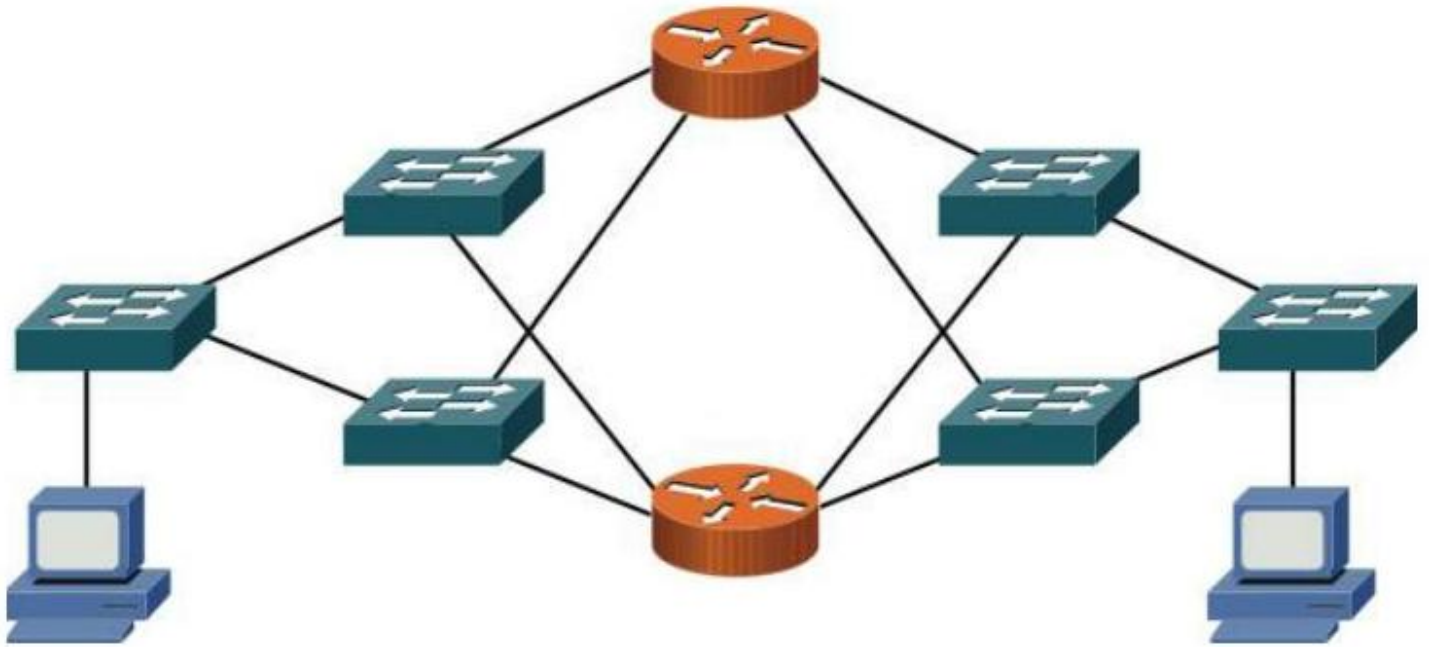
PC0 to PC3 and another targeted from PC1 to PC5.

https://www.geeksforgeeks.org/implementation-of-tree-topology-in-cisco/?ref=ml_lbp

Star and Mesh Topology



Fault-Tolerant for No Single Point of Failure



08

Network Protection Overview

A short preview before future security lectures



Why Protect a Network?

Security is part of network design, but detailed security comes later



Unauthorized access
users or devices that should not enter



Malicious traffic
harmful or suspicious communication



Service disruption
attacks or failures that stop access

Network Defense Tools

Examples from the source deck

- **Firewall**
- **IDS**
- **IPS**
- **NAC**
- **Content Filter**
- **Anti-DDoS**
- **Proxy**
- **Load Balancer**

Firewall

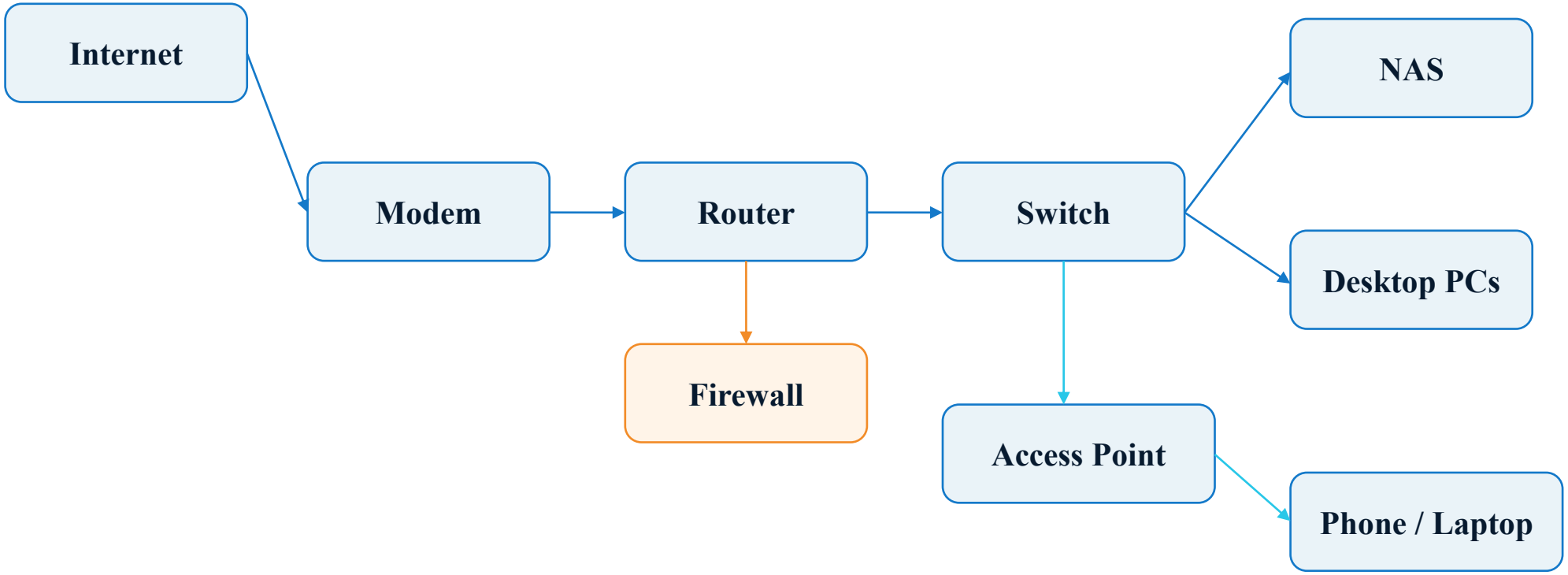
A boundary control between external and internal networks



Allowed traffic ✓ · Blocked traffic ✗

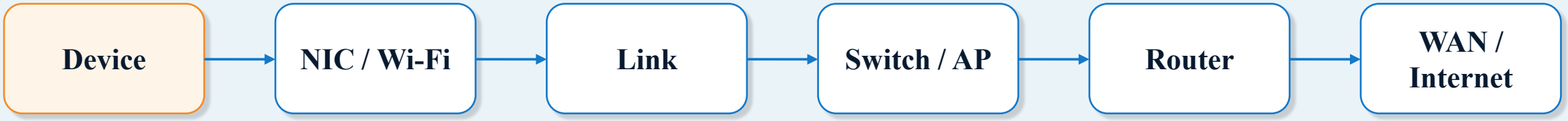
Putting It All Together

One small network combines devices, links, wireless, storage, and security



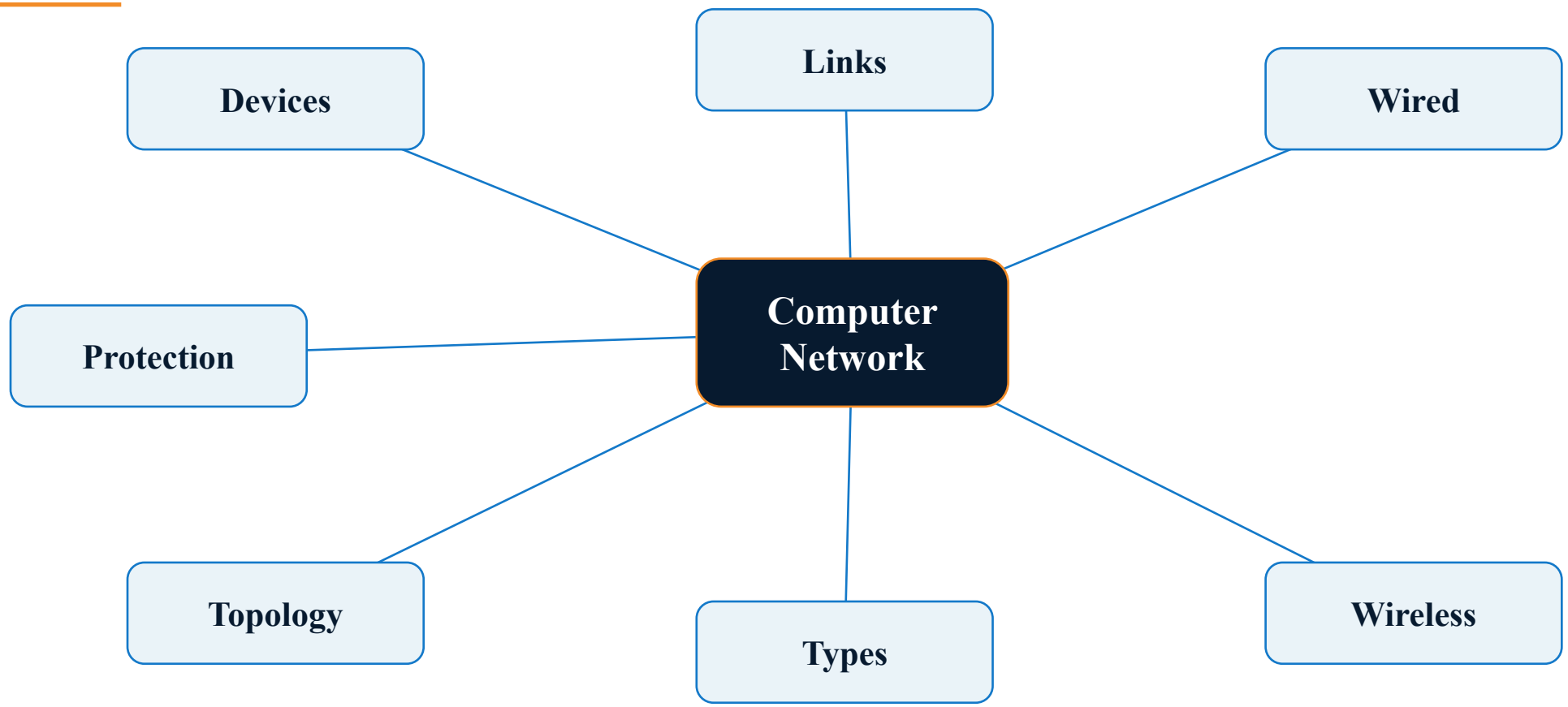
From Device to Internet

The end-to-end path of a basic network request



Concept Map

Computer Network as connected concepts



Key Takeaways

The core ideas to remember

- Networks connect devices through links.
- Devices have different roles: repeat, switch, route, access, protect.
- Links may be copper, fiber, or wireless.
- Network type depends on scale and purpose.
- Topology describes how nodes and links are arranged.

Mini Activity: Design a Small Office Network

Apply the whole lecture in one scenario

Requirements

- 8 desktop PCs
- Wi-Fi for mobile devices
- Internet connection
- shared storage

Questions

What devices are required?

Which links should be wired?

Which topology fits best?

Suggested Architecture

One valid design for the mini activity



This is not the only valid answer; explain your design decisions.

Before Next Week

Computer Network → Network Communication

- How do different devices understand each other?
- How are messages organized?
- How are packets transmitted through networks?



Thank You

Questions?